

12.164

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Question

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

Find the maximum number of linearly independent eigenvectors of $\mathbf{A}$.

Solution

Since $\mathbf{A}$ is a triangular matrix

$$\lambda = 1, 1, 1 \quad (1)$$

$$(\mathbf{A} - \mathbf{I})\mathbf{x} = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \mathbf{x} = 0, \quad \mathbf{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \quad (2)$$

$$\Rightarrow \begin{pmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad (3)$$

$$R_1 \rightarrow R_1 - R_2 \Rightarrow \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad (4)$$

$$z = 0, y = 0 \quad (5)$$

$$\mathbf{x} = \begin{pmatrix} x \\ 0 \\ 0 \end{pmatrix} = x \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad (6)$$

Maximum number of linearly independent eigenvectors = 1

C Code

```
#include <stdio.h>
#include <math.h>

int main() {
    // Matrix A
    double A[3][3] = {
        {1, 1, 1},
        {0, 1, 1},
        {0, 0, 1}
    };

    // Step 1: Since A is upper triangular, eigenvalues are
    //           diagonal elements
    double lambda[3];
    for (int i = 0; i < 3; i++) {
        lambda[i] = A[i][i];
    }

    printf(Eigenvalues:\n);
```

C Code

```
for (int i = 0; i < 3; i++) {  
    printf("%d = %.2f\n", i+1, lambda[i]);  
}  
  
// Step 2: All eigenvalues are equal (1), find eigenvectors  
for = 1  
// (A - I)  
double AI[3][3];  
for (int i = 0; i < 3; i++) {  
    for (int j = 0; j < 3; j++) {  
        AI[i][j] = A[i][j];  
        if (i == j) AI[i][j] -= 1.0;  
    }  
}  
  
printf("\nMatrix (A - I):\n");  
for (int i = 0; i < 3; i++) {
```

```
    for (int j = 0; j < 3; j++) {
        printf("%6.2f", AI[i][j]);
    }

    printf("\n");
}

// From manual solving:
// x3 = 0, x2 = 0, x1 = free 1 eigenvector
printf("\nEigenvector corresponding to  $\lambda = 1$ :\n");
printf("v1 = [1 0 0]^T\n");

printf("\nMaximum number of linearly independent eigenvectors  
= 1\n");
return 0;
}
```

Python Direct

```
import numpy as np

# Define matrix A
A = np.array([
    [1, 1, 1],
    [0, 1, 1],
    [0, 0, 1]
], dtype=float)

# Step 1: Eigenvalues
eigenvalues, _ = np.linalg.eig(A)
print(Eigenvalues:)
for i, val in enumerate(eigenvalues, start=1):
    print(f{i} = {val:.2f})

# Step 2: For eigenvalue = 1, compute (A - I)
I = np.eye(3)
AI = A - I
```



```
print(\nMatrix (A - I):)
print(AI)

# Step 3: Find null space to get eigenvectors
# (A - I)x = 0   eigenvectors for = 1
u, s, vh = np.linalg.svd(AI)
tol = 1e-10
null_mask = (s <= tol)
null_space = np.compress(null_mask, vh, axis=0)
```

```
print(\nEigenvectors corresponding to  = 1:)
if null_space.size == 0:
    print(No eigenvectors found.)
else:
    print(null_space.T)

# Step 4: Number of linearly independent eigenvectors
num_independent = null_space.shape[0]
print(f\nMaximum number of linearly independent eigenvectors = {
    num_independent})
```